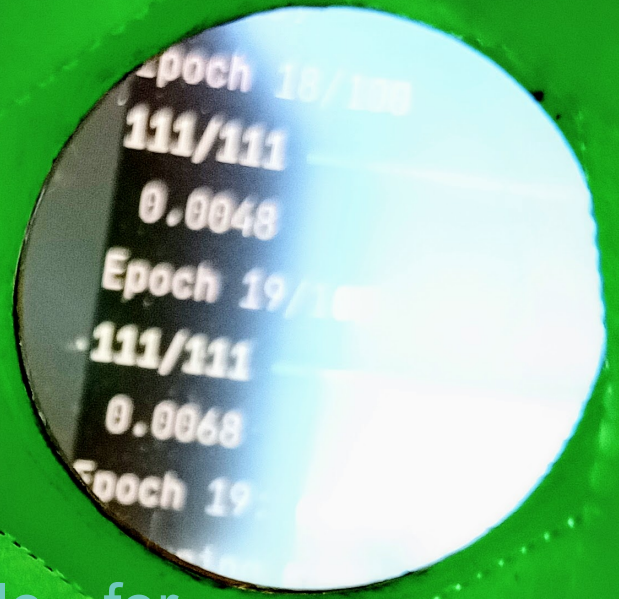


Lifelong Learning for a Kresling Origami Robot

A Plug-and-play Module for Supervised Learning of Target Adaptations

E. De Vroey



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by

E. De Vroey

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Student number: 4685156
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Thesis committee: Dr. ing. J. Kober TU Delft, supervisor
Dr. M. Wiertlewski TU Delft

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Preface

When asked numerous times over the past year what I was doing my Master's Thesis on, I can remember thinking every single time how lucky I was with the title of my topic. *"Lifelong Learning for a Kresling Origami Robot"* speaks to the imagination, and turns the person who asked the question—likely out of politeness—into a genuinely interested listener.

Upon finding this thesis topic amongst the other options, I was surprised that it was still available: Origami meets machine learning, and you get to use your *arts-and-crafts*-skills while designing and building your very own robot—surely the list of available options had not been updated yet. To my great relief, Professor Kober assured me that it was still free, and we sat down to discuss the prospects. I had recently finished an internship where I gained some hands-on experience in implementing machine learning by myself, and I like crafting and tinkering, so this topic seemed cut-out for me. That said, I would not have considered myself a "maker". I never worked with microcontrollers before, nor soldered my own circuit-boards together; I hardly really understood the basic principle of a breadboard. Thirteen months later, I can say that I do not regret choosing this topic.

I would therefore like to thank Professor Jens Kober for his encouragement and support throughout the entire process. Often finding myself at an impasse, Professor Kober would offer some valuable insight or alternative solution to look into, completely freeing me up to *get moving* again. Probably also indebted to Professor Kober for that same reason are my parents, whom I really need to thank for continually putting up with my repeated statements that *"yes, I really am nearly finished now"*. Their worries of me becoming an eternal student might have been a great deal worse, however, were it not for the kind people at the TU Delft Science Centre, who allowed me to access their MakerSpace. Without their lasercutter, the—to my own estimation—200 origami structures that I fabricated throughout the process would have likely taken months to cut out and construct.

Then there are my friends: I spent nearly the entirety of my thesis working in the Echo building, and without the assurance that Bram, Oscar, Thomas, Niels, and Kato would be there as well, all going through the same tough process, it would have been a lot harder to get up and go there every morning. Our so-called "Echo crew" was always ready to go for a spontaneous coffee break, to lunch together, or to grab a drink after a long day of work. On that note, I think I can spare a line of text to extend my gratitude to the volunteers at Architecture's faculty bar De BouwPub, who were always willing to provide said drinks. My housemates, both old and new, Tim, Frederique, Emiel, Tine, Loes, Maxim, Rune, Sien, and Nelson, I would like to thank for making each day more than just *work*. May the Bronx Blues Brothers soon have their musical breakthrough. Finally, I would like to thank Arnaud, who volunteered to proofread this attempt at a coherent work of scientific research—a thankless task which I hope he does not regret accepting.

With that said, I hope that you enjoy *"Lifelong Learning for a Kresling Origami Robot"* as much as I enjoyed the whole process of turning this concept from a mere title into a physical robot, and finally, into this document.

E. De Vroey
Delft, October 2024

Contents

Preface	iii
Contents	v
Lifelong Learning for a Kresling Origami Robot	1
I Introduction	1
II Related Work	2
II-A Soft Robotics	2
II-B Lifelong Learning	2
II-C Kresling Origami	2
III Methodology	3
IV Illustrative Problem	4
IV-A A case where the controller is too aggressive	4
IV-B A case where the controller is under-responsive	5
V A Kresling Origami Robot	6
V-A Mechanical Details	6
V-B Software Implementation	7
VI Experimental Setup	7
VI-A Sign Conventions and Unit Clarification	7
VI-B Performance Metrics	8
VII Experiments and Results	8
VII-A Varying the PID Controller	8
VII-B Varying the System Dynamics	10
VII-C Crawling Gait	12
VIII Discussion	12
VIII-A Key Insights	13
VIII-B Limitations	13
VIII-C Validity of the Results	14
IX Conclusion	15
Appendix A: KOS Design and Template	16
Appendix B: Software implementation	16
B-A Nominal Control on the Arduino	16
B-B Target Adaptation on the External Computer	16
Appendix C: Extended Test Results	17
Appendix D: KOS Reference Recordings	23

